

Maternal and Offspring Genetic Risk of Type 2 Diabetes and Offspring Birthweight Among African Ancestry Populations

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Objectives: Maternal genetic risk of type 2 diabetes (T2D) can influence offspring birthweight through shared offspring genetic risk and by altering intrauterine glycemic status. The aim of this study was to estimate the independent effects of maternal and offspring genetic risk scores (GRSs) of T2D on offspring birthweight and the extent to which intrauterine glycemic traits mediate the effect of maternal GRSs on offspring birthweight.

Design: The study involved 949 mother–offspring pairs of African ancestry from the Hyperglycemia Adverse Pregnancy Outcome study. GRSs of T2D were calculated separately for mothers and offspring as the weighted sum of 91 T2D risk alleles identified in a genome-wide association study meta-analysis in African Americans. Linear regression models were fit to estimate changes in birthweight by quartiles of GRSs. Mediation analysis was implemented to estimate the direct and indirect effects of maternal GRS on offspring birthweight through cord blood C-peptide and maternal fasting and postchallenge glucose levels.

Results: Maternal and offspring GRSs were independently and differentially associated with offspring birthweight. Changes (95% CI) in birthweight across increasing quartiles of maternal GRSs were 0 g (reference), 83.1 g (6.5, 159.6), 103.1 g (26.0, 180.2), and 92.7 g (12.6, 172.8) (P trend = 0.041) and those of offspring GRSs were 0 (reference), −92.0 g (−169.2, −14.9), −64.9 g (−142.4, 12.6), and 2.0 g (−77.8, 81.7) (P trend = 0.032). Cord blood C-peptide mediated the effect of maternal GRS on offspring birthweight, whereas maternal postchallenge glucose levels showed additive effects with maternal GRS on birthweight.

Conclusions: Maternal and offspring GRSs of T2D were independently and differentially associated with offspring birthweight. (*J Clin Endocrinol Metab* 104: 5032–5042, 2019)

Abnormal birthweight is an established risk factor of neonatal mortality and morbidity, as well as a predictor of later-life cardiometabolic diseases, including type 2 diabetes (T2D) (1, 2). Complex interactions of maternal and offspring genetic and environmental risk factors influence birthweight. It is estimated that

maternal and offspring genotypes account for 3% to 22% and 24% to 69% of the variations in birthweight, respectively (3–6), indicating the importance of studying both maternal and offspring genetic contributions for a comprehensive understanding of the genetic architecture of birthweight.

Maternal genetic contributions to offspring birthweight can be attributed to maternally inherited offspring risk alleles or the influence of maternal genotypes on components of the intrauterine environment, such as the glycemic status during pregnancy (7, 8). Maternal glycemic status plays a major role in determining fetal growth, as demonstrated by positive associations of maternal fasting blood glucose levels, prepregnancy diabetes, and gestational diabetes with offspring birthweight (9–13). Maternal glucose level stimulates the secretion of fetal insulin (11), which is an important fetal growth factor (14). A recent genome-wide association study (GWAS) meta-analysis has identified 10 maternal genetic loci associated with offspring birthweight; 7 of the loci are reported to exert their influence through altering intrauterine environment, as demonstrated by their associations with T2D, fasting glucose, fasting insulin and proinsulin, blood pressure, lipid levels, and coronary artery disease (7).

In the offspring genome, there is an overall inverse genetic correlation between birthweight and adulthood risk of T2D (4). However, across individual loci, T2D single-nucleotide polymorphisms (SNPs) with shared effect on birthweight exhibit heterogeneous associations; whereas T2D risk alleles in some loci are associated with increased birthweight, those in other loci are associated with decreased birthweight. For example, Horikoshi *et al.* (4) tested 84 T2D risk loci for associations with birthweight and identified that T2D risk alleles in *ADCY5*, *CDKAL1*, and *HHEX-IDE* genes were strongly associated with lower birthweight, whereas risk alleles in *ANK1* and *MTNR1B* genes were associated with higher birthweight. These findings are reinforced by observational studies reporting a “U-shaped” relationship between birthweight and adulthood risk of T2D, where both low and high birthweight have been associated with increased risk of T2D in adulthood (15). Whether, and to what extent, offspring genetic risk of T2D contributes to offspring birthweight independently of maternal genetic risk and *vice versa* are not known.

To date, 403 T2D genetic variants at 243 loci have been identified by GWAS meta-analyses in European ancestry populations (16, 17). In contrast, 104 loci have been identified in African ancestry populations (18), a group often underrepresented in GWASs of complex traits (19) despite having a higher prevalence of diabetes (21.8%) (20) and low birthweight (13.2%) compared to the general population (21). In the current study, we sought to estimate the independent effects of maternal and offspring genetic risk of T2D on offspring birthweight by aggregating individual risk effects of recently identified T2D SNPs in African Americans into a single genetic risk score (GRS). We also estimated the extent to

which intrauterine glycemic traits mediate the effect of maternal GRS on offspring birthweight.

Materials and Methods

Study population

The current study included genotype data of mother–offspring pairs from the Hyperglycemia Adverse Pregnancy Outcome (HAPO) study (July 2000 to April 2006). The details of the study were described previously (9, 22). Briefly, women aged ≥ 18 years with a viable pregnancy of < 31 weeks were eligible for enrollment (22). Eligible women were scheduled to have a 75-g 2-hour oral glucose tolerance test (OGTT) between 24 and 32 weeks of gestation (target time of testing, 28 weeks). During the OGTT visit, standardized questionnaires were administered to collect maternal anthropometric and demographic information. Medical history was abstracted from medical records review. Body mass index (BMI) was calculated as weight in kilograms divided by the square of height in meters at the time of OGTT and categorized into three groups: underweight and normal, < 25.0 ; overweight, 25.0 to < 30 ; and obese, ≥ 30 .

Cord blood C-peptide and maternal fasting and 75-g postchallenge glucose levels at 1 and 2 hours were used as proxies for intrauterine glycemic status. Maternal fasting and postchallenge glucose levels were measured at the center laboratories using enzymatic methods with extensive quality control, as described previously (23). Cord blood was collected at delivery for the measurement of serum C-peptide and to extract fetal DNA. Serum C-peptide was measured at the central laboratory of the HAPO study by immunoassay (AutoDELFIA, PerkinElmer) (23, 24).

Gestational age was determined from the date of the last menstrual period or using ultrasound. Neonatal anthropometric measurements were obtained using standardized methods and equipment within the first 72 hours of delivery. Birthweight was obtained using a calibrated electronic scale to the nearest 10 g.

Maternal and offspring DNA were genotyped at the genome-wide level using Illumina Human1M-Duo BeadChip. Quality control of the genotypes was performed as reported previously (25). Genotypes were imputed with the Michigan Imputation Server (26) implementing Eagle2 (27) for haplotype phasing, followed by Minimac2 (28) for imputing nontyped SNPs with 1000 Genomes phase 3 data (29). Genetic ancestry components were determined using a model-based estimation of ancestry implemented in the program ADMIXTURE (30).

A total of 1345 Afro-Caribbean participants were included in the original study, of whom genotype data were collected from 1157 mothers and 1130 offspring (976 mother–offspring pairs). This study included data from 949 mother–offspring pairs after excluding participants with a proportion of African ancestry $< 10\%$ ($n = 10$) to minimize bias due to population structure (31), missing offspring birthweight ($n = 13$), birthweight > 5000 g ($n = 2$), and offspring with a duplicate study identification number ($n = 2$).

GRS of T2D

We selected a total of 104 SNPs associated with T2D based on a recent GWAS meta-analysis in African Americans (18). We excluded 11 SNPs because of low imputation quality score ($r^2 < 0.3$) and an additional 2 SNPs because of missing genotype

information. A weighted GRS was calculated based on 91 T2D SNPs (32) separately for each mother and offspring as the sum of the dosage of individual risk allele multiplied by the corresponding effect size reported in the GWAS meta-analysis (18).

Statistical analysis

Demographic and prenatal data of the study participants were summarized with descriptive statistics. Missing data on maternal years of education ($n = 9$) were imputed with multiple imputation by chained equation using information of maternal age, education, height, weight, and parity under the assumption of missing at random (33).

The effect of maternal and offspring GRSs on birthweight was estimated using multivariable linear regression models for increasing quartiles of GRSs compared with the lowest quartile. Models were adjusted for *a priori* selected covariates: the proportion of maternal or offspring African ancestry (continuous), maternal age (continuous), years of education (continuous), BMI at the time of OGTT (underweight or normal weight, overweight, obese), parity (nulliparous, parous), gestational age at delivery (continuous), and offspring sex (male, female). Both maternal and offspring GRSs were included in the same model to estimate the independent effect of one from the other. We tested for trend over increasing quartiles of GRSs. We also examined the pattern of associations for maternal and offspring GRSs with birthweight using penalized cubic splines, adjusting for the same set of covariates. We also compared mean birthweight by above and below the median values of cord blood C-peptide and maternal fasting and postchallenge glucose levels at the time of OGTT across quartiles of maternal GRSs using a nonparametric test for trend (34).

For association between maternal GRS and offspring birthweight, we also tested for mediation and interaction by cord blood C-peptide and maternal fasting and postchallenge glucose levels at 1 and 2 hours. Additional exploratory analyses were performed using maternal BMI, glycated hemoglobin (HbA1c), systolic and diastolic blood pressure, and cord blood plasma glucose levels as potential mediators. To conduct mediation analysis, we ran two sets of models: the model for birthweight (outcome model) and the model for respective glycemic trait (mediator model). The mediator and outcome models were combined to estimate the direct, indirect, and total effects of maternal GRS on birthweight across increasing quartiles of maternal GRSs (35). The model for birthweight was adjusted for maternal and offspring GRSs, the proportion of maternal African ancestry, maternal age, years of education, BMI, parity, infant sex, and gestational age at birth, in addition to the respective mediating variable (*i.e.*, glycemic trait). The model for respective glycemic trait was adjusted for maternal and offspring GRSs, maternal age, education, parity, BMI, infant sex, and gestational age at blood collection. We also tested for interaction by adding an interaction term between maternal GRS and the respective glycemic trait in the model for birthweight. Direct and indirect effects were averaged across all individuals. CIs were estimated using the delta method (36). Mediation analysis assumes that there are no unmeasured confoundings of the exposure–mediator, exposure–outcome, and mediator–outcome relationships, and there is no unmeasured confounder of the mediator–outcome relationship that itself is affected by the exposure (37). All statistical tests were two-sided and were performed using Stata version 14.2 (StataCorp).

Results

The distribution of maternal and offspring characteristics is presented in Table 1. The mean ($\pm$ SD) of birthweight and gestational age at delivery was 3216.2 ($\pm$ 442.9) g and 39.8 ($\pm$ 1.2) weeks, respectively. More than 60% of the mothers were either overweight (BMI $\geq$ 25.0 to $<$ 30) or obese (BMI $\geq$ 30) at the time of OGTT. The medians (interquartile range) of maternal fasting, 75-g 1-hour and 2-hour postchallenge glucose levels were 79.2 (9.0), 118.8 (32.4), and 106.2 (25.2) mg/dL, respectively. Maternal fasting plasma glucose was moderately correlated with postchallenge glucose levels at 1 hour (Spearman $\rho = 0.35$; 95% CI: 0.29, 0.40) and 2 hours (Spearman $\rho = 0.33$; 95% CI: 0.27, 0.39). Cord blood C-peptide showed weak correlations with maternal fasting (Spearman $\rho = 0.29$; 95% CI, 0.23, 0.35) and postchallenge glucose levels at 1 and 2 hours (Spearman $\rho = 0.19$; 95% CI, 0.13, 0.25).

We observed independent associations of maternal and offspring GRSs with offspring birthweight (Table 2). Higher maternal GRS was significantly associated with increased birthweight after adjusting for offspring GRS; changes (95% CI) in mean birthweight across increasing quartiles of maternal GRSs were 0 g (reference), 83.1 g (6.5, 159.6), 103.1 g (26.0, 180.2), and 92.7 g (12.6, 172.8), respectively (P trend = 0.041). In contrast, offspring GRS was inversely associated with birthweight; changes (95% CI) in mean birthweight across increasing quartiles of offspring GRS were 0 g (reference), -92.0 g (-169.2 , -14.9), -64.9 g (-142.4 , 12.6), and 2.0 g (-77.8 , 81.7), respectively (P trend = 0.032). Our results suggested that the associations of maternal and offspring GRSs with offspring birthweight were not linear (Fig. 1; Table 2). Specifically, offspring GRS exhibited a U-shaped relationship with birthweight, where both low and high offspring GRSs were associated with higher offspring birthweight.

Additionally, maternal GRS appeared to have an additive effect with maternal glycemic traits in relationship to birthweight (Fig. 2) (32). Across quartiles of maternal GRS, we observed significant trends of increasing mean birthweight by maternal fasting and postchallenge glucose levels at 1 and 2 hours as well as by cord blood C-peptide level. Alternatively, among women with postchallenge glucose levels below the median, we observed a significant trend of increasing mean birthweight by quartile of maternal GRS.

Mediation analysis revealed that the effect of maternal GRS on birthweight was mediated through cord blood C-peptide level (Table 3). For example, changes (95% CI) in mean birthweight mediated through cord blood C-peptide across increasing quartiles of maternal

Table 1. Distribution of Selected Characteristics Among Study Participants

Characteristics	n (%)	Mean $\pm$ SD or Median (IQR)
Maternal age, y		25.5 $\pm$ 5.6
Birthweight, g		3216.2 $\pm$ 442.9
Gestational age at birth, wk		39.8 $\pm$ 1.2
Maternal years of education		12.7 $\pm$ 1.4
Maternal BMI at the time of OGTT		
15.3 to <25.0	369 (38.9)	
25.0 to <30.0	281 (29.6)	
30.0 to 62.9	299 (31.5)	
Parity, nulliparous	488 (51.4)	
Offspring sex, female	461 (48.6)	
History of any hypertensive disorder	78 (8.2)	
Maternal GRS		115.3 $\pm$ 5.9
Q1		107.8 $\pm$ 2.9
Q2		113.4 $\pm$ 1.1
Q3		117.1 $\pm$ 1.1
Q4		122.9 $\pm$ 3.0
Offspring GRS		
Q1		90.06 $\pm$ 2.4
Q2		94.45 $\pm$ 1.2
Q3		97.7 $\pm$ 1.3
Q4		102.1 $\pm$ 2.3
Proportion of African ancestry of mother		0.92 (0.10)
Proportion of African ancestry of offspring		0.89 (0.11)
Maternal fasting plasma glucose, mg/dL		79.2 (9.0)
Maternal 75-g 1-h plasma glucose, mg/dL		118.8 (34.4)
Maternal 75-g 2-h plasma glucose, mg/dL		106.2 (25.2)
Cord blood C-peptide, μ g/L		0.8 (0.6)

Missing data were reported for maternal years of education ($n = 9$). Q1, Q2, Q3, and Q4 denote first, second, third, and fourth quartiles, respectively. Abbreviation: IQR, interquartile range.

GRSs were 0 g (reference), 26.7 g (5.7, 47.6), 9.1 g (−11.7, 29.9), and 24.7 g (2.1, 47.2), respectively, whereas changes in mean birthweight through pathways other than cord blood C-peptide were 0 g (reference), 53.6 g (−20.4, 127.6), 91.1 g (15.3, 166.9), and

65.1 g (−12.5, 142.8), respectively. Maternal fasting and postchallenge glucose levels measured at the time of OGTT did not mediate the effect of maternal GRS on offspring birthweight.

Discussion

In the current study of 949 mother–offspring pairs of African ancestry, we found that maternal and offspring GRSs of T2D were independently and differentially associated with offspring birthweight. Specifically, higher maternal GRS was associated with increased offspring birthweight; part of the effect was mediated through cord blood C-peptide level, highlighting fetal insulin-mediated growth. Our results also suggested that maternal GRS of T2D and postchallenge glucose levels have an additive effect on offspring birthweight, where higher mean birthweight was observed when both maternal GRS and postchallenge glucose levels were higher. In contrast, offspring GRS showed a U-shaped relationship with birthweight. Because offspring GRS reflects genetic susceptibility to future risk of T2D, our findings suggest that genetic risk factor may partly explain the well-known U-shaped relationship between birthweight and adulthood risk of T2D (38).

Table 2. Effects of Maternal and Offspring Genetic Risk for T2D on Offspring Birthweight

Genetic Risk for T2D	Coefficient (95% CI)	P Value
Maternal GRS		
Q1 (96.6–111.5)	Reference	
Q2 (111.5–115.1)	83.1 (6.5, 159.6)	0.033
Q3 (115.2–119.1)	103.1 (26.0, 180.2)	0.009
Q4 (119.2–133.5)	92.7 (12.6, 172.8)	0.023
P trend		0.041
Offspring GRS		
Q1 (95.4–111.1)	Reference	
Q2 (111.1–115.1)	−92.0 (−169.2, −14.9)	0.019
Q3 (115.1–119.2)	−64.9 (−142.4, 12.6)	0.101
Q4 (119.2–131.7)	2.0 (−77.8, 81.7)	0.962
P trend		0.032

Models were adjusted for maternal age, years of education, parity, BMI, offspring sex, gestational age at birth, and corresponding maternal or offspring GRSs and the proportion of maternal or offspring African ancestry. Bold type indicates statistically significant associations ($P < 0.05$). Q1, Q2, Q3, and Q4 denote first, second, third, and fourth quartiles, respectively.

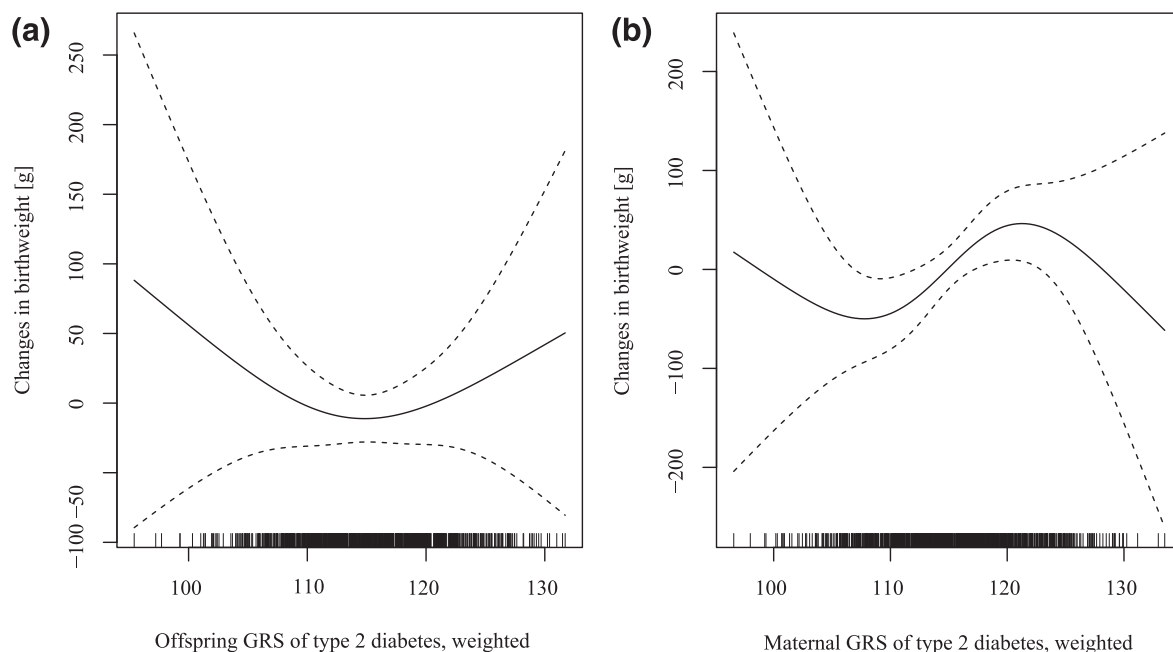


Figure 1. Pattern of associations for (a) maternal and (b) offspring GRSs of T2D with offspring birthweight (g) using natural cubic spline models. Models were adjusted for maternal age, years of education, parity, BMI, offspring sex, gestational age at birth, and corresponding maternal or offspring GRS and the proportion of African ancestry.

Studies on maternal GRS of T2D and birthweight among African ancestry populations are sparse. However, our observed association is consistent with a previous study in European populations reporting that maternal GRS based on a subset of glucose-raising alleles

was positively associated with offspring birthweight (7). Using maternal GRS of T2D as an instrumental variable, a recent Mendelian randomization study in European populations reported that maternal T2D was causally associated with higher offspring birthweight independently

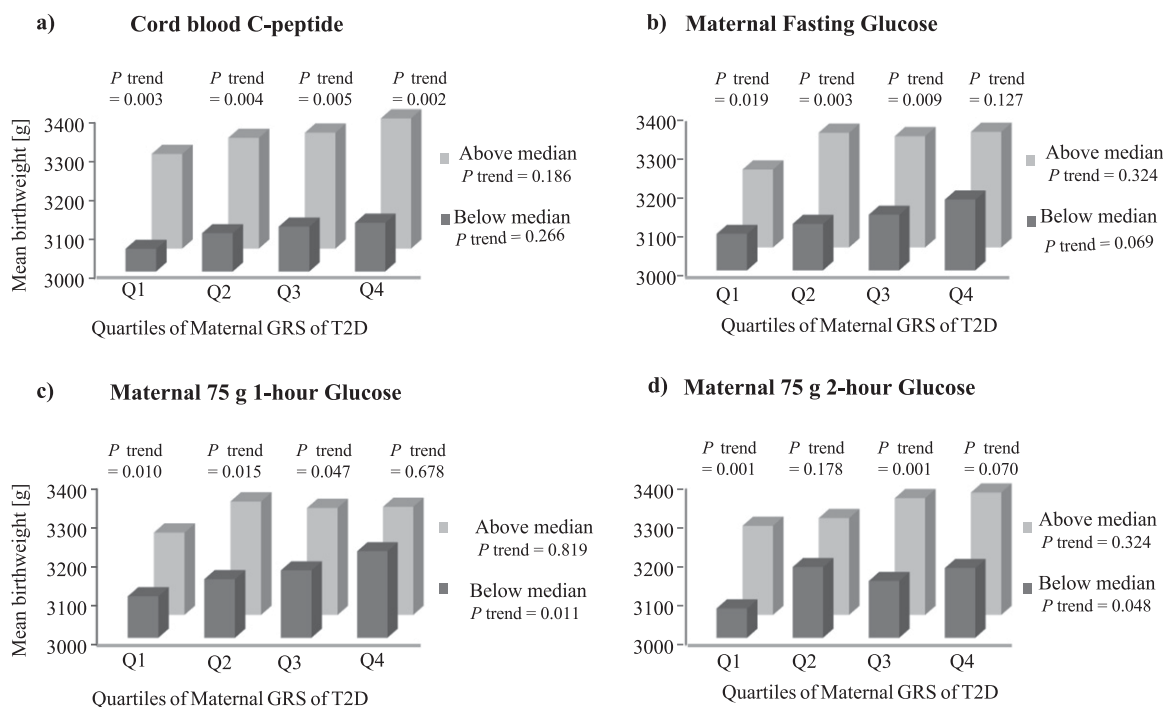


Figure 2. Combined effects of maternal GRS of T2D and intrauterine glycemic traits on offspring birthweight. (a–d) Bar chart showing mean birthweight (g) across (a) combined maternal GRS and offspring C-peptide, (b) maternal fasting glucose, and postchallenge glucose levels at (c) 1 h and (d) 2 h. Black bars represent mean offspring birthweight when glycemic markers were below the corresponding median value, whereas gray bars represent mean offspring birthweight when glycemic markers were above the corresponding median value. Corresponding *P* values for the linear trend across quartiles of maternal GRS and categories of glycemic markers are also presented. Q1, Q2, Q3, and Q4 denote first, second, third, and fourth quartiles, respectively.

Table 3. Indirect, Direct, and Total Effects of Maternal GRS on Offspring Birthweight Considering Respective Intrauterine Glycemic Trait as a Mediator

Glycemic Markers	GRS of T2D	Indirect Effects		Direct Effects		Total Effects	
		Coefficient (95% CI)	P Value	Coefficient (95% CI)	P Value	Coefficient (95% CI)	P Value
Natural log of fasting plasma glucose	Q1	Ref.		Ref.		Ref.	
	Q2	1.6 (−6.2, 9.3)	0.692	80.2 (2.1, 158.3)	0.001	81.8 (3.2, 160.3)	0.041
	Q3	0.5 (−7.4, 8.4)	0.892	101.3 (24.6, 178.0)	0.044	101.9 (24.5, 179.2)	0.010
	Q4	7.6 (−1.5, 16.8)	0.102	83.9 (5.7, 162.1)	0.010	91.5 (12.6, 170.5)	0.023
Natural log of 75-g 1-h plasma glucose	Q1	Ref.		Ref.		Ref.	
	Q2	2.7 (−6.2, 11.6)	0.556	79.1 (1.6, 156.6)	0.046	81.8 (3.3, 160.3)	0.041
	Q3	10.1 (−0.2, 20.2)	0.054	92.1 (15.8, 168.4)	0.018	102.1 (24.7, 179.5)	0.010
	Q4	10.8 (−0.4, 22.1)	0.059	81.2 (2.3, 160.1)	0.044	92.0 (12.9, 171.1)	0.023
Natural log of 75-g 2-h plasma glucose	Q1	Ref.		Ref.		Ref.	
	Q2	1.1 (−8.2, 10.3)	0.821	81.2 (3.8, 158.6)	0.040	82.3 (3.8, 160.7)	0.040
	Q3	4.5 (−5.3, 14.2)	0.373	98.1 (22.9, 173.3)	0.011	102.5 (25.2, 179.8)	0.009
	Q4	7.7 (−3.2, 18.6)	0.165	84.6 (6.6, 162.5)	0.033	92.3 (13.4, 171.2)	0.022
Natural log of cord blood C-peptide	Q1	Ref.		Ref.		Ref.	
	Q2	26.7 (5.7, 47.6)	0.012	53.6 (−20.4, 127.6)	0.156	80.3 (2.5, 158.1)	0.043
	Q3	9.1 (−11.7, 29.9)	0.391	91.1 (15.3, 166.9)	0.019	100.2 (21.6, 178.8)	0.012
	Q4	24.7 (2.1, 47.2)	0.032	65.1 (−12.5, 142.8)	0.100	89.8 (7.9, 171.7)	0.032

Models for birthweight (outcome model) were adjusted for maternal age, years of education, BMI at the time of OGTT, parity, proportion of maternal African ancestry, offspring GRS and sex, and gestational age at birth. Models for glycemic traits (mediator models) were adjusted for maternal age, years of education, BMI at the time of OGTT, parity, offspring GRS and sex, and gestational age at blood collection. Bold type indicates statistically significant associations ($P < 0.05$). Q1, Q2, Q3, and Q4 denote first, second, third, and fourth quartiles, respectively.

Abbreviation: Ref., reference.

of the offspring genotype (14). This finding is further reinforced by observational studies showing a positive association between maternal diabetes and offspring birthweight (39, 40).

Maternal genetic susceptibility to diabetes can influence offspring birthweight via maternally inherited offspring risk alleles or altering the intrauterine environment, particularly maternal glycemic status during pregnancy (8). A recent GWAS meta-analysis of 86,577 women of European ancestry identified maternal SNPs at 10 loci (*MTNR1B*, *HMG2*, *SH2B3*, *KCNAB1*, *L3MBTL3*, *GCK*, *EBF1*, *TCF7L2*, *ACTL9*, and *CYP3A7*) that were significantly associated with offspring birthweight; at least 7 of the loci represented maternal intrauterine effects, primarily acting upon maternal glycemic status, vascular function, or placental transfer of nutrients (7). Specifically, maternal variants associated with offspring birthweight at three loci (*MTNR1B*, *GCK*, and *TCF7L2*) were known to be associated with fasting glucose and T2D susceptibility (41, 42), with the glucose raising alleles associated with higher offspring birthweight independently of the fetal genotypes (7).

Maternal glucose level influences birthweight primarily by stimulating the secretion of offspring insulin, which is an important fetal growth factor (13, 14, 43). Consistent with this finding, our results revealed that the effect of maternal GRS on birthweight was mediated through cord blood C-peptide, suggesting fetal insulin-mediated growth. In contrast, maternal fasting and

postchallenge glucose levels or HbA1c level (data not shown) measured at the time of OGTT did not mediate the effect of maternal GRS on offspring birthweight. HbA1c level can be inaccurate in African populations because among this subgroup, there is a higher prevalence of sickle cell trait, one type of hemoglobinopathies reported to underestimate past glycemia (44). Maternal fasting and postchallenge glucose levels were also weakly correlated with cord blood C-peptide, suggesting that maternal glycemic traits measured in late second trimester may not adequately capture fetal insulin-mediated growth. Fetal pancreatic β cells are recognizable from late first trimester, although insulin secretion from fetal pancreas is not glucose sensitive until approximately the 28th week of gestation (45). Therefore, it is possible that maternal glycemic markers measured in early to middle pregnancy may not adequately capture fetal insulin-mediated growth.

Maternal genetic risk of T2D can also influence offspring birthweight via altering body adiposity status, placental vascular function, and placental transfer of nutrients to the fetus (14, 46, 47). In fact, several T2D risk loci have also been identified to have shared effects on gestational diabetes (48, 49), obesity (50, 51), pregnancy weight gain (52), and blood lipids (53); each of these factors can independently influence offspring birthweight (14). A recent study by Shrestha *et al.* (54), involving African ancestry populations of the HAPO study, the same source population of the current study,

found that offspring GRS of BMI was inversely associated with birthweight, whereas maternal GRS of BMI modified the effect of maternal prepregnancy BMI on birthweight. GRS of BMI in Shrestha *et al.*'s (54) study and GRS of T2D in the current study were constructed using different sets of SNPs that only overlapped at two loci (data not shown); therefore, it is unlikely that our observed association between maternal GRS of T2D and offspring birthweight was influenced by SNPs that have a shared effect on T2D and obesity. Further exploratory analyses indicated that neither maternal BMI at the time of OGTT nor maternal hypertensive status, systolic, or diastolic blood pressures mediated the effect of maternal GRS of T2D on offspring birthweight (32).

Studies that examined offspring GRS of T2D on birthweight among African ancestry populations are sparse. A recent multiethnic GWAS meta-analysis of birthweight reported inverse genetic correlation between birthweight and elevated risk of T2D (4). A significant inverse association between offspring GRS estimated based on 25 T2D loci risk alleles and birthweight was also observed in the Danish Inter99 study, although the associations between individual risk allele and birthweight were heterogeneous (55). T2D risk alleles associated with lower birthweight were those typically characterized by reduced insulin processing and secretion without detectable changes in fasting glucose (7). In line with this finding, observational studies have linked paternal diabetes with lower offspring birthweight (39). These findings are consistent with the “fetal insulin hypothesis” that posits that genetically determined insulin resistance or deficiency leads to low insulin-mediated intrauterine fetal growth as well as insulin resistance later in life (43). In contrast, T2D risk alleles associated with higher birthweight were those typically characterized by hyperglycemia (*MTNR1B*, *GCK*), thus increasing fetal insulin-mediated growth (7). Using GRS of T2D, we observed a U-shaped relationship between offspring GRS and birthweight after adjusting for maternal GRS, recapitulating the well-known U-shaped relationship between birthweight and offspring's future risk of diabetes observed in observational studies (38).

In line with our findings, a recent study by Hughes *et al.* (11) identified that fetal birthweight GRS estimated based on 60 birthweight SNPs (4) was positively associated with birthweight independently of maternal fasting plasma glucose, where fetal GRS and maternal fasting plasma glucose showed an additive effect. Some T2D risk loci have been reported to have a shared effect on birthweight. For example, of the 60 birthweight SNPs used to construct fetal birthweight GRS by Hughes *et al.* (11), 7 SNPs were located in genes also reported to have associations with T2D (17, 18). However, it is unlikely

that our observed association between offspring GRS of T2D and birthweight was driven by the seven SNPs having shared effect on T2D and birthweight, as GRS based on the seven SNPs did not show significant associations with birthweight whereas our observed associations between maternal GRS and offspring birthweight remained consistent after excluding the seven SNPs (data not shown).

Our study has important limitations, which need to be considered when interpreting the findings. We did not have data on fetal insulin level, which is an important determinant of fetal growth. In the HAPO study, nearly 15% of the blood samples collected from umbilical veins showed detectable hemolysis (23). Hemolysis is known to interfere with the level of insulin but not that of C-peptide, which is secreted from pancreatic β -cells along with insulin in equimolar levels (22). Hence, cord blood C-peptide level serves as a good surrogate for fetal insulin level as an index of offspring β -cell function. We did not use cord blood glucose level in our analysis because the level of glucose falls rapidly in cord blood when there is a delay in processing due to the high glycolytic rate of erythrocytes of the newborn, as was observed for samples collected during the earlier phase of the HAPO study (23).

Our study has several strengths. By incorporating genetic information from maternal–offspring pairs and adjusting for the GRS during analysis, we were able to estimate the effect of distinct maternal and offspring GRS on birthweight. Moreover, GRS was constructed based on 91 T2D SNPs identified in African ancestry populations, a group often underrepresented in genetic studies (19), despite having a higher prevalence of diabetes (20) and low birthweight (21). However, out of the 91 T2D SNPs included in our study's GRSs, 52 SNPs were identical with or LD proxies ($r^2 \geq 0.4$) of SNPs associated with T2D in European ancestry populations (exact match or $r^2 \geq 0.80 = 30$ SNPs; $r^2 \geq 0.60 - 0.79 = 16$ SNPs; $r^2 \geq 0.40 - 0.59 = 6$ SNPs) (16, 32), suggesting some degree of transethnic genetic risk conferred by shared genetic loci. Whether the findings of our study are generalizable to non-African descent populations warrants further study.

In conclusion, we observed independent and differential associations of maternal and offspring GRSs of T2D on offspring birthweight. Higher maternal GRS was associated with increased birthweight, where part of the effect was mediated through cord blood C-peptide, suggesting fetal insulin-mediated growth. Furthermore, we observed a U-shaped relationship between offspring GRS of T2D and birthweight, concordant with the well-known U-shaped relationship between birthweight and adulthood risk of T2D in observational studies. A

comprehensive understanding of the contribution of maternal and offspring genetic risk of T2D on offspring birthweight may help to tailor antenatal care to minimize the risk of adverse birth outcomes.

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